

ECO387C - Fall 2024

Problem Set 8

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

1 Endowment economy with a cooler

There are two periods: period 0 and period 1, and one good. There is only one type of agent, a household. Assume that the total mass of agents is $N = 1$. In period 0, each agent seeks to maximize the expected lifetime utility.

$$\max \mathbb{E}_0 \sum_{t=0}^1 \beta^t u(c_t)$$

In addition to the risk-free asset (a) with an interest rate of $1 + r$, the agent has another means of saving: cooler (x). The agent can save consumption goods in the cooler now and take them out for consumption in the next period. Thus, the agent is subject to the following budget constraints:

$$\begin{aligned} c_0 + x + a &= y_0 \\ c_1 &= y_1 + x + (1 + r)a \\ x &\geq 0 \end{aligned}$$

1. Write down the agent's maximization problem as a Lagrangian. Find the Euler equations.
2. Suppose $\beta = 1$ and $u(c) = \log(c)$. Find the equilibrium consumption and interest rates in this economy considering the following two endowment profiles:
 - $(y_0, y_1) = (4, 3)$
 - $(y_0, y_1) = (3, 4)$
3. Repeat the previous question with $u(c) = -\frac{1}{2}(c - \bar{c})^2$ where $\bar{c} > 0$ is some large number representing a satiation point.

2 Endowment economy without a cooler

Consider the same optimization problem of the agent but suppose now that there is no cooler.

Part 1: Without uncertainty Consider the endowment profile $(y_0, y_1) = (4, 3)$

1. Write down the agent's maximization problem as a Lagrangian. Find the Euler equations.
2. Suppose $\beta = 1$ and $u(c) = \log(c)$. Find the equilibrium consumption and interest rates $(1 + r_{\log,1}^*)$ in this economy.
3. Suppose $\beta = 1$ and $u(c) = -\frac{1}{2}(c - \bar{c})^2$ where $\bar{c} \gg 0$ is some large number representing a satiation point. Find the equilibrium consumption and interest rates $(1 + r_{\text{quad},1}^*)$ in this economy.

Part 2: With uncertainty Suppose now that the period 1 endowment is uncertain while the period 0 endowment remains the same at 4. Specifically, the period 1 endowment will be either high $y_1^h = 5$ or low $y_1^l = 1$. The probability of getting a high endowment is $1/2$.

1. Write down the agent's maximization problem as a Lagrangian. Find the Euler equation.
2. Suppose $\beta = 1$ and $u(c) = \log(c)$. Find the equilibrium consumption and interest rates $(1 + r_{\log,2}^*)$ in this economy. How does $(1 + r_{\log,2}^*)$ compare with $(1 + r_{\log,1}^*)$?
3. Suppose $\beta = 1$ and $u(c) = -\frac{1}{2}(c - \bar{c})^2$ where $\bar{c} \gg 0$ is some large number representing a satiation point. Find the equilibrium consumption and interest rates $(1 + r_{\text{quad},2}^*)$ in this economy. How does $(1 + r_{\text{quad},2}^*)$ compare with $(1 + r_{\text{quad},1}^*)$?